

**BRAC University**  
**CSE341: Microprocessor**  
**Quiz#3**

Duration : 20 Minutes

Total Marks : 20

**ID:**

**Name:**

**Sec:**

1.

| Address | 00083h | 00084h | 00085h | 00086h | 00087h | 1B5C0h | 1B5C1h | 1B5C2h |
|---------|--------|--------|--------|--------|--------|--------|--------|--------|
| Data    | 9Ch    | A0h    | 15h    | 02h    | 1Ah    | 5Bh    | 6Eh    | 78h    |

- a. Explain if the interrupt caused by the INT 33 instruction is maskable or not. Provide your reasoning. [1]
  - b. If the Intel 8086 encounters the INT 33 instruction, then calculate the starting address of the corresponding ISR. Also, what will be the first 2-byte hex instruction that will be fetched to the instruction queue from the ISR? [1 + 2 + 1]
  - c. Explain in detail all the steps on how an 8086 responds to a signal received on the INTR pin. [4]
2. A student designed a new PIC called 'PIC 2.0' which takes 20 interrupt requests via its IRR0 –IRR19 pins. The size of its Interrupt Mask register (IMR), Interrupts Service Register (ISR), and Interrupt Request Register (IRR) is 20 bits. It also has 4 Cascading (CAS) pins.
- a. Assume the value of ISR0 - ISR4 is 10000, the value of IMR0 - IMR4 is 10101, and the value of IRR0 - IRR4 is 01110. Now, explain the order in which the interrupts IR0-IR4 will be serviced. Assume IR0 has the highest priority and IR19 has the least priority. [3]
  - b. Calculate the maximum number of interrupts that a master PIC 2.0 can handle in cascade mode. Show necessary calculations. [2]
  - c. Due to a design flaw in the PIC 2.0, the student noticed that the master PIC 2.0 could not communicate with all the connected 20 slave PIC 2.0s. Explain how you would correct the flaw in the design so that the master can communicate with all the 20 slave PIC 2.0s. [2]
  - d. Explain 2 possible methods of stopping the servicing of any interrupts between the master and the 8086 without disconnecting the wires or disabling the INTR / INT / INTA' pins. [2]